

Mingyun Kang

Researcher in Construction Management & Automation

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Education

Seoul National University of Science and Technology

Bachelor in Architectural Engineering, School of Architecture

Seoul, Republic of Korea

Mar 2017 – Feb 2024

- Double Major: Business Administration
- GPA : 3.8 / 4.5

Seoul National University of Science and Technology

Master in Architectural Engineering (Integrated Bachelor-Master Program)

Seoul, Republic of Korea

Mar 2024 – Aug 2025

- GPA : 4.22 / 4.5
- Thesis : Computer Vision-based Adhesion Quality Inspection Model for Exterior Insulation and Finishing System
- Advisor : Prof. Taehoon Kim

Research Experience

Research Assistant

CONICT Lab, Seoul National University of Science and Technology

Seoul, Republic of Korea

Dec 2021 – Aug 2025

Conducted research on computer vision-based automated construction supervision, 3D reconstruction, and human-robot collaboration at construction sites under Prof. Taehoon Kim.

- Developed a computer vision-based adhesion quality inspection model for Exterior Insulation and Finishing System (EIFS)
- Published 3 SCIE journal papers and 2 SCOPUS conference papers as first/co-author
- Filed 5 domestic patent applications related to construction automation and vision-based inspection
- Participated in 3 national R&D projects funded by NRF and the Ministry of Land, Infrastructure and Transport (MOLIT)

Post-Master's Researcher

Korea Institute of Civil Engineering and Building Technology (KICT)

Goyang-si, Republic of Korea

Nov 2025 – present

Awards

Best Paper Award (최우수학술상)

May 2023

Awarded at the Spring Conference 2023 for the paper: A Basic Study on Image-based Numerical Measurement Model for Construction Sites

Korean Institute of Building Construction (KIBC)

Best Paper Award (최우수학술상)

May 2024

Awarded at the Spring Conference 2024 for the paper: A Basic Study on Rebar Spacing Measurement Method Based on Single Image Information

Korean Institute of Building Construction (KIBC)

Excellence Paper Award (우수학술상)

May 2025

Awarded at the Spring Conference 2025 for the paper: A Basic Study on Image Analysis-based Crackdown on Illegal Buildings on Rooftops

Korean Institute of Building Construction (KIBC)

Outstanding Thesis Award (우수논문상)

Aug 2025

Awarded for the Master's thesis: Computer Vision-based Adhesion Quality Inspection Model for Exterior Insulation and Finishing System

Seoul National University of Science and Technology

Publications

Computer Vision-Based Adhesion Quality Inspection Model for Exterior Insulation and Finishing System

Applied Sciences, 15(1), 125. (SCIE)

Mingyun Kang, Sebeen Yoon, Taehoon Kim

Approach to Enhancing Panoramic Segmentation in Indoor Construction Sites Based on a Perspective Image Segmentation Foundation Model

Applied Sciences, 15(9), 4875. (SCIE)

Juho Han, Sebeen Yoon, Mingyun Kang, Taehoon Kim

Automated Vision-based Location Monitoring for Jack Support

KSCE Journal of Civil Engineering, 100374. (SCIE)

Sebeen Yoon, Mingyun Kang, Minhyung Kim, Taehoon Kim

Instance Segmentation of Exterior Insulation Finishing System using Synthetic Datasets

ISARC Vol. 41, pp. 1176-1181. IAARC Publications. (SCOPUS)

Mingyun Kang, Sebeen Yoon, Juho Han, Sanghyeon Na, Taehoon Kim

Comparative Study of Structure from Motion on Construction Site

ISARC Vol. 42, pp. 1395-1400. IAARC Publications. (SCOPUS)

Mingyun Kang, Sangmin Lee, Sebeen Yoon, Taehoon Kim

Skills

Computer Vision

Languages

Korean

Native speaker

English

Proficient — TOEIC Speaking 160, Advanced Low (Valid: Nov. 2023 – Nov. 2025)

Interests

Architectural Engineering & AI

Certificates

Architectural Engineer License (건축기사)

June 2025

TOEIC Speaking — Advanced Low (Score 160)

Nov 2023

Projects

Digital Mapping-based Platform Research Laboratory for Healthy Construction Lifecycle

June 2022 – Feb 2025

National R&D project exploring digital mapping-based platforms for the full construction lifecycle. Focused on computer vision applications for automated construction supervision.

- Vision-based adhesive application quality inspection for EIFS
- Tile quantity counting via computer vision

Development of Digital-based Architectural Supervision and Construction Automation Robot Technology

Apr 2022 – Dec 2026

Large-scale national R&D project (KRW 24,490M) funded by MOLIT aimed at developing digital architectural supervision systems and construction automation robots.

- Instance segmentation of insulation finishing systems using synthetic datasets
- Panoramic image segmentation for indoor construction sites

Development of Human-Robot Collaboration Technology and Integrated Operation System for High-Altitude Work at Construction Sites

Apr 2025 – Dec 2027

Ongoing national R&D project (KRW 17,000M) funded by MOLIT developing XR-based human-robot collaboration platforms for multipurpose high-altitude construction work.

- XR-based human-robot collaboration interface
- Integrated robot operation system for construction sites

References

Professor Taehoon Kim